

Questions		Answers	Marks
1	(a)	Repeated readings with error of different magnitude and sign. OR Repeated readings with scatter of readings about a mean value. By plotting a graph and drawing a line of best fit for the points/can be reduced by averaging	B1 B1
	(b)	Units of $\frac{kr^3(p_1-p_2)}{l} \sqrt{\frac{M}{RT}} = \frac{m^3 Pa}{m} \sqrt{\frac{kg mol^{-1}}{J K^{-1} mol^{-1} K}}$ $= kgms^{-2} \sqrt{\frac{kg s^2}{kgm^2}}$ $= kg s^{-1}$	
	(c) (i)	Density = m/V m = density x V = density x $\pi(D^2/4) L = (8.96) \times \pi(6.0^2/4) L = 7093.46 \text{ g} = 7.09346 \text{ kg}$ $\Delta m/m = 2\Delta D/D + \Delta L/L = (2(0.1)/(6.0) + (0.1/28.0)) = 0.0369$ $\Delta m = m(2\Delta D/D + \Delta L/L) = (7093.46)(0.0369) = 0.262 \text{ kg}$ 7.1 +/- 0.3 kg (Correct s.f.)	M1 M1 A1 A1
	(ii)	Same measurements, but smaller absolute uncertainties, so percentage uncertainty is smaller	B1

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2	(a)	apply equation of motion $v = u + at$ at vertically, we get	